

Basic Immunology

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December, 2022

Immunologic concepts

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- Despite continual exposure to microbes, we become *ill rarely*
 - How does the body defend itself?
 - During infection, how does the body eliminate the invader and cure itself?
 - How does the body defend itself from subsequent infections after primary infection?
 - Is there any means to get protection from a pathogen encountered for the first time?

Definition

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- Immunology grew out of the observation
 - *“individuals who had recovered from certain infectious diseases were thereafter protected from the disease”*
- Immunity
 - Derived from the Latin term ‘*immunis*’
 - *Immunis* means “exempt” → Immunity: the state of protection from infectious disease
- Immunology
 - The study of all aspects of host defense against infection including the adverse consequences of an immune responses

Historical Perspectives

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Thucydides in 430 BC in Athens

- earliest written reference to the phenomenon of immunity
- wrote that **only those who had recovered from plague could nurse the sick** because they would not contract the disease a second time
- Although early societies recognized the phenomenon of immunity,
 - It took almost 2000 years before the concept was successfully converted into medically effective practice

Historical Perspectives

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Chinese and Turks in the 15th century

- The 1st recorded attempts to induce immunity deliberately
- They were attempting to prevent **smallpox**,
 - **fatal** in about 30% of cases
 - leaves survivors **disfigured** for life
- Dried crusts derived from smallpox pustules were either inhaled into the nostrils or inserted into small cuts in the skin (**variolation**)

Historical Perspectives...

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Edward Jenner, in 1798



- made a giant advance in the deliberate development of immunity
- **Observation:** Milkmaids who had contracted the mild disease cowpox were subsequently immune to smallpox
- **Hypothesis:** introducing fluid from a cowpox pustule into people might protect them from smallpox
- **Proof**
 - inoculated an 8 years old boy with fluid from a cowpox pustule → intentionally infected the child with smallpox → the child didn't develop smallpox
- Jenner called his procedure vaccination

Historical Perspectives...

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Louis Pasteur (1822–1895)

- next major advance: the induction of immunity to cholera
- Louis Pasteur had succeeded in growing etiology of fowl cholera
- Confirmed this by injecting it into chickens → developed **fatal cholera**
- After returning from a vacation, he and colleagues resumed their experiments, injecting some chickens with an **old culture**
- The chickens became ill, but to Pasteur's surprise, **they recovered**
- Pasteur then grew a fresh culture of the bacterium
- his supply of fresh chickens was limited, and therefore he used a **mixture** of previously injected and unexposed chickens
- Unexpectedly, only the **fresh chickens died** and the **others protected**

Historical Perspectives...

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Louis Pasteur (1822–1895)

- Pasteur hypothesized and later showed that
 - aging had weakened the virulence of the pathogen
 - such a weakened or attenuated strain could be administered to provide immunity against the disease.
- He called this attenuated strain a **vaccine** (in Latin vacca means “cow”)
- Pasteur extended these findings to other diseases
 - demonstrated that it was possible to attenuate a pathogen and administer the attenuated strain as a vaccine

Historical Perspectives...

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Louis Pasteur (1822–1895)

- In 1881, Pasteur first vaccinated one group of sheep with *Bacillus anthracis* attenuated by heat treatment
- He then challenged the vaccinated sheep, along with some unvaccinated sheep, with a virulent culture of the anthrax bacillus
- All the vaccinated sheep lived and all the unvaccinated animals died
- These experiments marked the beginnings of the discipline of immunology
- In 1885, Pasteur administered his first vaccine to a human, a young boy who had been bitten repeatedly by a rabid dog





Cells and organs of the immune system

Cells & organs of the immune system...Introduction

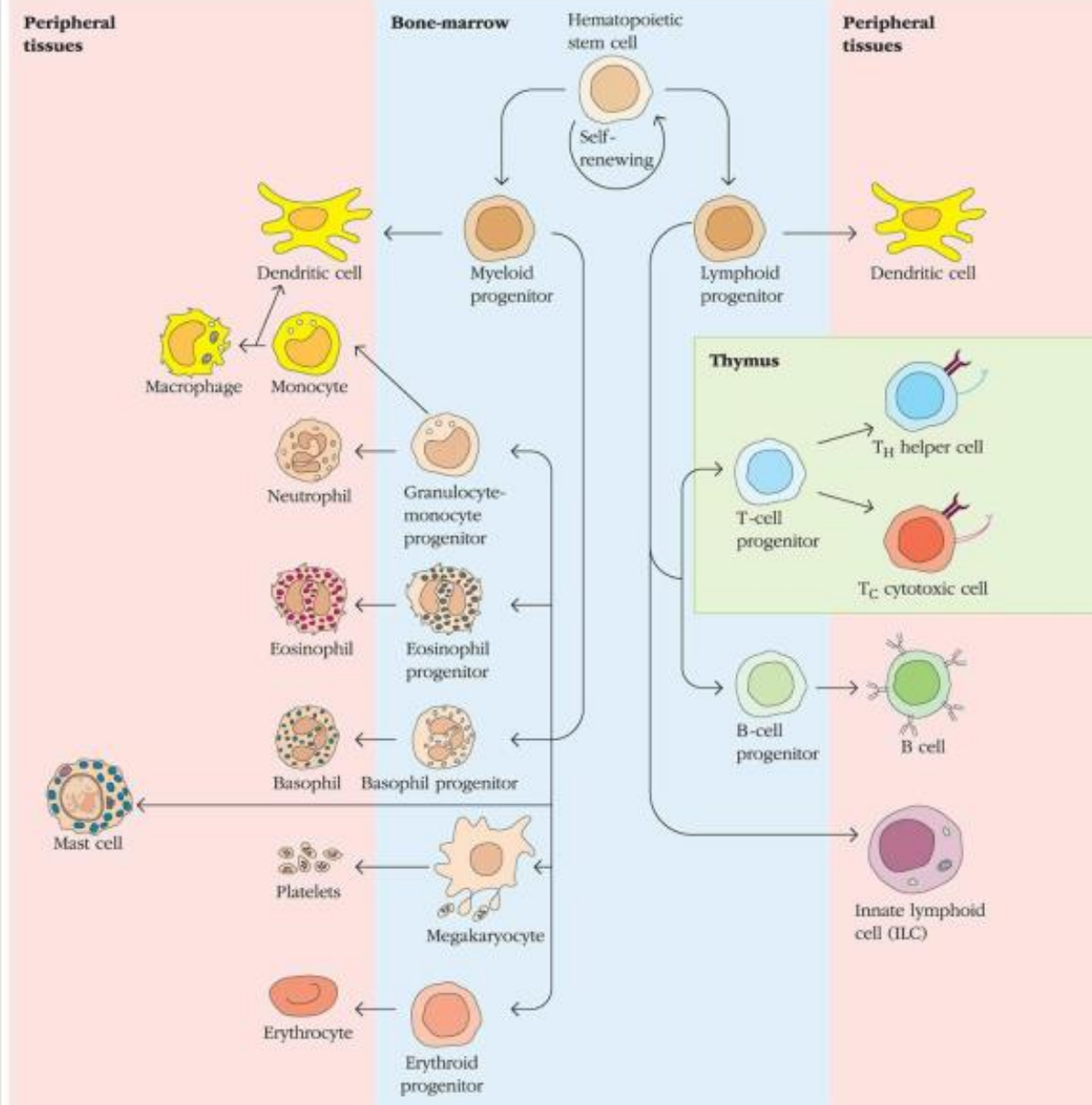
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- **Hematopoiesis**
 - process by which hematopoietic stem cells (HSCs) differentiate into mature blood cells
- **Stem cells are defined by two capacities**
 - a. ability to regenerate or “self-renew”
 - b. ability to differentiate into diverse cell types
- **Hematopoietic stem cell are source of all of our blood cells**

Cells & organs of the immune system...

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- All blood cells originate from the same progenitor or precursor cells called the pluripotent hematopoietic stem cells (PHSC) in the bone marrow
- PHSC divide to produce lymphoid progenitor and myeloid progenitor cells
- lymphoid progenitor gives rise to T and B lymphocytes, Natural Killer (NK) cells
- The common myeloid progenitor is the precursor of
 - Macrophages, granulocytes, mast cells, and dendritic cells of the innate immune system
 - megakaryocytes and red blood cells



Punt, *Kuby Immunology*, 8e, © 2018 W. H. Freeman and Company

FIGURE 2-1 Hematopoiesis. Self-renewing hematopoietic stem cells give rise to lymphoid and myeloid progenitors. Most immune cells mature in the bone marrow and then travel to peripheral organs via the blood. Some, including mast cells and macrophages, undergo further maturation outside the bone marrow. T cells develop to maturity in the thymus.

Immune Cells of the Myeloid Lineage

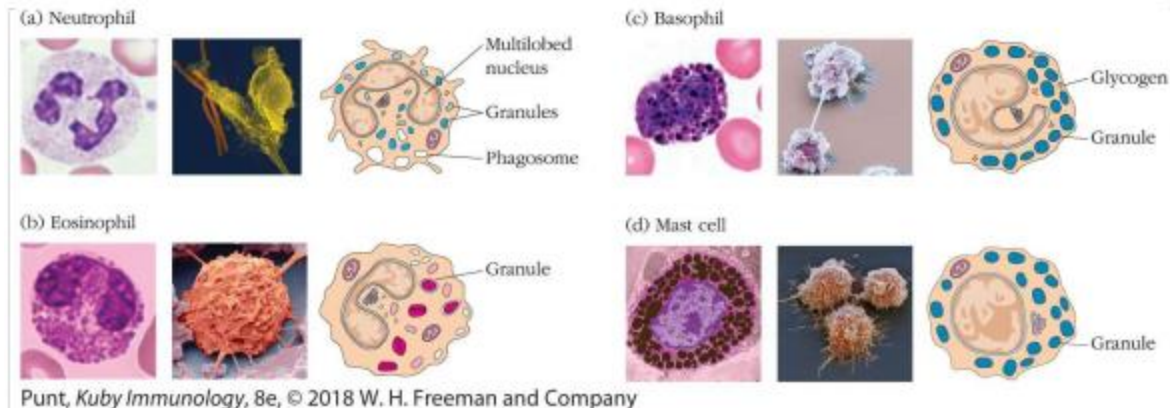
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- **Are the First Responders to Infection**
- **include granulocytes, monocytes, and macrophages**
- **The white blood cells within this lineage are innate immune cells**
 - **Respond rapidly to the invasion of a pathogen**
 - **Communicate the presence of an insult to cells of the lymphoid lineage**
 - **contribute to inflammatory diseases (asthma and allergy)**

Granulocytes

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- first responders during an immune response
- fall into four main categories: neutrophils, eosinophils, basophils, and mast cells
- All granulocytes have multilobed nuclei
- Granulocyte subtypes differ by the staining characteristics of their cytoplasmic granules



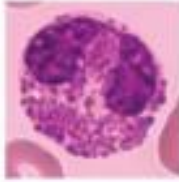
Granulocytes...



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Neutrophils

- Constitute the majority (50% to 70%) of circulating leukocytes in adult humans
- are the most numerous and most important cellular component of the innate immune response
- are the first to arrive at a site of inflammation
- Functions
 - phagocytose (engulf) bacteria
 - secrete a range of proteins (*e.g. Defensins, lysozyme*) that have antimicrobial effects
- Neutrophils are the main cellular components of pus



Granulocytes...

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Eosinophils

- constitute 1%–3% of circulating leukocytes
- Contain granules that stain a brilliant pink in standard H&E staining protocols
- Important in coordinating our defense against multicellular parasitic organisms, including helminths (parasitic worms)
- Eosinophils cluster around invading worms, and damage their membranes by releasing contents of their granules
- Contributors to asthma and allergy symptoms

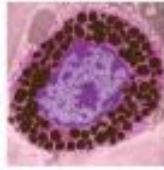


Granulocytes...

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Basophils

- are relatively rare in the circulation (<1%)
- Are non-phagocytic with large basophilic granules that stain **blue** in standard H&E staining
- Bind circulating antibody/antigen complexes → release contents of their granules; e.g. Histamine
- Histamine
 - increases blood vessel permeability and smooth muscle activity
 - allows immune cells access to a site of infection
- Function
 - response to parasites, particularly helminths (parasitic worms)
 - Contribute to allergy



Granulocytes...

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Mast cells (<1%)

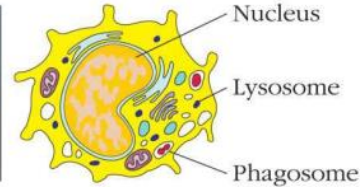
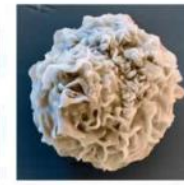
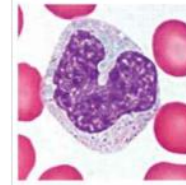
- Like circulating basophils, these cells have large numbers of cytoplasmic granules
- are released from the bone marrow into the blood as undifferentiated cells
- mature after they leave the blood for a wide variety of tissues
 - skin, connective tissues of various organs, and mucosal epithelial tissue of the respiratory, genitourinary, and digestive tracts
- **Function**
 - combating parasitic worms
 - contribute to allergies
- Basophils and mast cells share many features but have distinct origins and functions

Myeloid Antigen-Presenting Cells

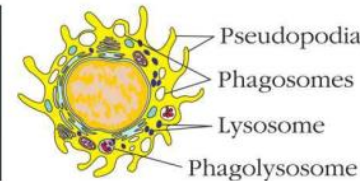
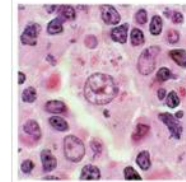
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- Include monocytes, macrophages, and dendritic cells
- professional antigen-presenting cells (pAPCs)
- pAPC form important cellular bridges between innate & adaptive immune systems
- They become activated after making contact with a pathogen at the site of infection
- They communicate this encounter to T lymphocytes by displaying peptides from the pathogen

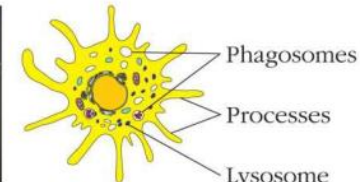
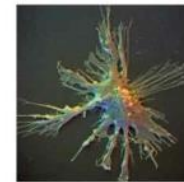
(a) Monocyte



(b) Macrophage



(c) Dendritic cell

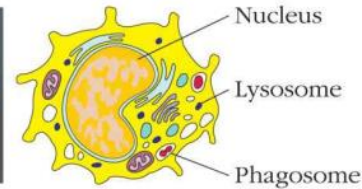
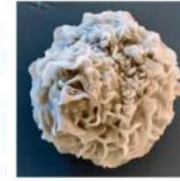
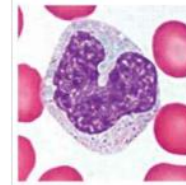


Myeloid Antigen-Presenting Cells...

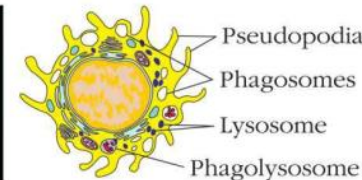
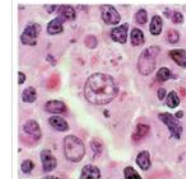
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- Professional APCs exhibit three major activities when they encounter pathogens:
 - 1) They secrete proteins that attract and activate other immune cells
 - 2) They internalize pathogens via phagocytosis, digest pathogenic proteins into peptides, and then present these peptide antigens on their membrane surfaces via MHC class II
 - 3) They upregulate co-stimulatory molecules required for optimal activation of helper T cells

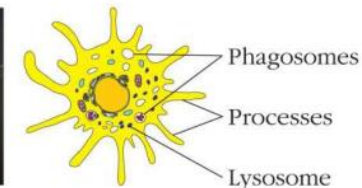
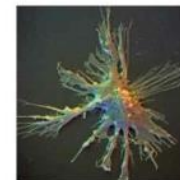
(a) Monocyte



(b) Macrophage



(c) Dendritic cell

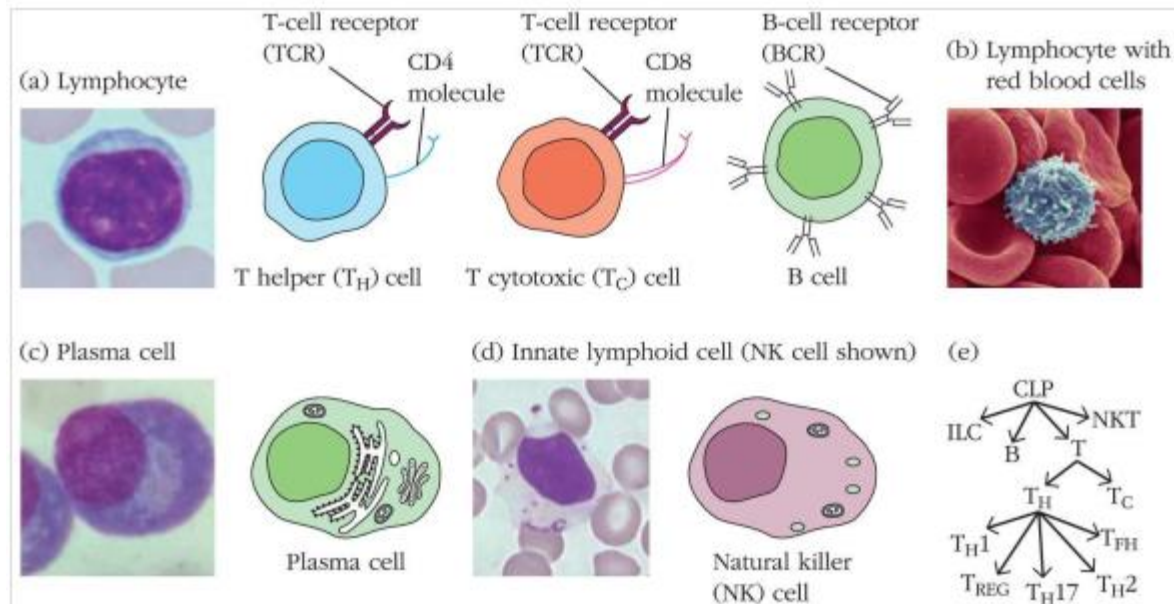
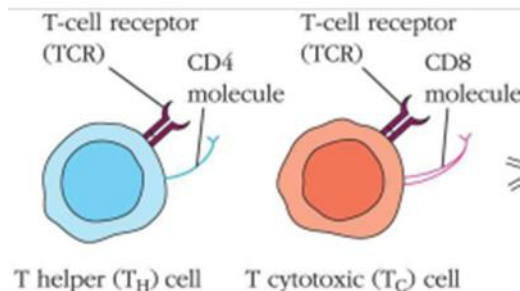


lymphocytes

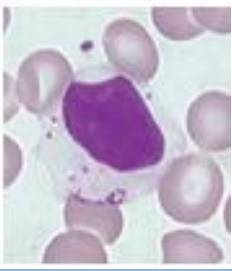
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- Represent 20% to 40% of circulating white blood cells and 99% of cells in the lymph
- Lymphocytes are broadly subdivided into three major populations on the basis of functional and phenotypic differences: B cells, T cells, and innate lymphoid cells (ILCs)

• T-cells are divided into T_H and T_C cells



Natural Killer cells



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- Large lymphocytes with distinctive cytotoxic granules
- Constitute 5%–10% of peripheral blood lymphocytes
- important in the innate immune defense against intracellular pathogens
- NK cells express a limited set of invariant receptors
 - Recognize indicators of infection, cancer, or damage expressed by other cells
- NK cells respond immediately to appropriate stimuli
 - Release effector proteins from preformed secretory granules that kill altered cells by inducing apoptosis
- NK cells express a membrane receptor for the Fc region of IgG
 - Antibody dependent cell-mediated cytotoxicity (ADCC)

Organs of the Immune System

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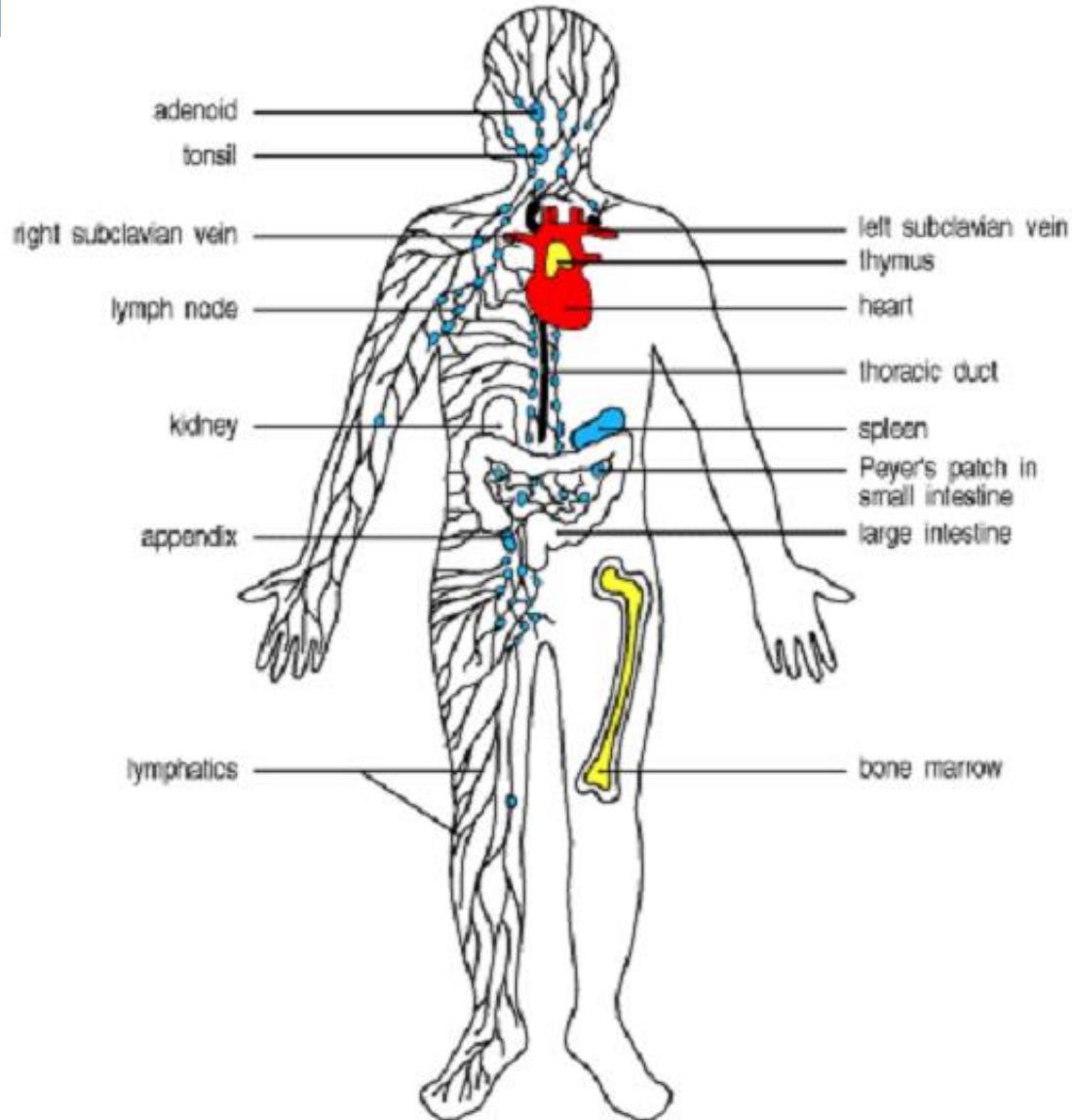
Lymphoid organs

- organized tissues containing large numbers of lymphocytes in a framework of non-lymphoid cells
- interactions of lymphocytes with non-lymphoid cells in these organs are important for
 - lymphocyte development
 - initiation of adaptive immune responses

Organs of the Immune System...

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- Lymphoid organs can be divided broadly into
1. Central or primary lymphoid organs
 2. Peripheral or secondary lymphoid organs



Primary/central lymphoid organs

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- Are organs where lymphocytes are generated and matured
- The primary lymphoid organs are the bone marrow and the thymus
- While B-cells originate and mature in the bone marrow/spleen, T-cells mature in the thymus
- Matured lymphocytes become committed to a particular antigenic specificity

Maturation of T and B lymphocytes

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- During thymocytes development an enormous diversity of T-cell receptors are generated by a random process
- T cells that cannot recognize self-MHC complexes and those that **react with self-antigen–MHC with high affinity die** by apoptosis
- B cells with **self-reactive** antibody receptors are also negatively selected in the bone marrow
- More than 95% of all thymocytes and 90% of B cells die by apoptosis without ever reaching maturity

Maturation of T and B lymphocytes...

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- Once matured, both B and T lymphocytes enter the bloodstream, from which they migrate to the peripheral lymphoid organs

Secondary/peripheral Lymphoid Organs

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- lymph nodes, spleen, and mucosal-associated lymphoid tissue (MALT) (Peyer's patches, the tonsils, adenoid and the appendix)
- antigen is carried from sites of infection, primarily within macrophages and dendritic cells
- antigen and lymphocytes encounter each other in the peripheral lymphoid organs
- Adaptive immune responses are initiated in these tissues

Antigens

Antigens

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- Any substance (usually foreign) that binds specifically to an antibody or a T-cell receptor
- Antigens are usually protein or carbohydrate in nature
- May be found as
 - free single molecules or
 - as parts of larger structures (e.g., expressed on the surface of an infectious agent)

Antigens...

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- Some antigens are very small and simple
- Others are large and complex
 - containing many different sites that can be individually identified by T-or B-cell receptors
 - Each such individual part of an antigen that can be distinctly identified by the immune system is called an **epitope or antigenic determinant**
 - The more complex the molecule and the greater the number of epitopes it displays, the more potent it is as an immunogen

Antigens...

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- Depending on their ability to stimulate an immune response, antigens can be immunogens or haptens

i. Immunogen

- can, by itself, both stimulate an adaptive immune response and subsequently serve as a target of that response

ii. Hapten

- cannot, by itself, stimulate an immune response
- if attached to a larger immunogenic molecule (a “carrier”), responses can be stimulated against both the carrier and the hapten

Factors That Influence Immunogenicity

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- a. **Foreignness:** nonself substances are immunogenic
- b. **Route of administration**
- c. **Dose**
 - the larger the dose of an immunogen, the greater the immune response is
 - Low dose (less immunogenic), Very high dose (immune tolerance)
- d. **Molecular size**
 - 10,000 Da (minimum molecular weight), 100,000 Da (best immunogens)
- e. **Chemical composition and molecular complexity**
 - copolymers composed of different amino acids or sugars are usually more immunogenic than homopolymers of their constituents
 - Proteins are stronger immunogenic than carbohydrates
- **Adjuvants:** substances when administered together with an immunogen enhance the response against it

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Overview of immune system

Overview of immune system

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Immune system

- Evolved to protect the host from invading infectious agents
- involves many organs, molecules, cells, and pathways
- Consists two interconnected systems
 - a. Innate**
 - b. Adaptive/acquired**
- These two systems collaborate to protect the body against foreign invaders

Overview of immune system...

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Innate immunity (summary)

- The first line of defense against infection
- Most of its components are present before the onset of infection
- Not specific to a particular pathogen (broad reactivity)
- Uniform in all members of a species
- Most microorganisms are readily cleared within few days by defense mechanisms of the innate immune system before they activate the adaptive immune system

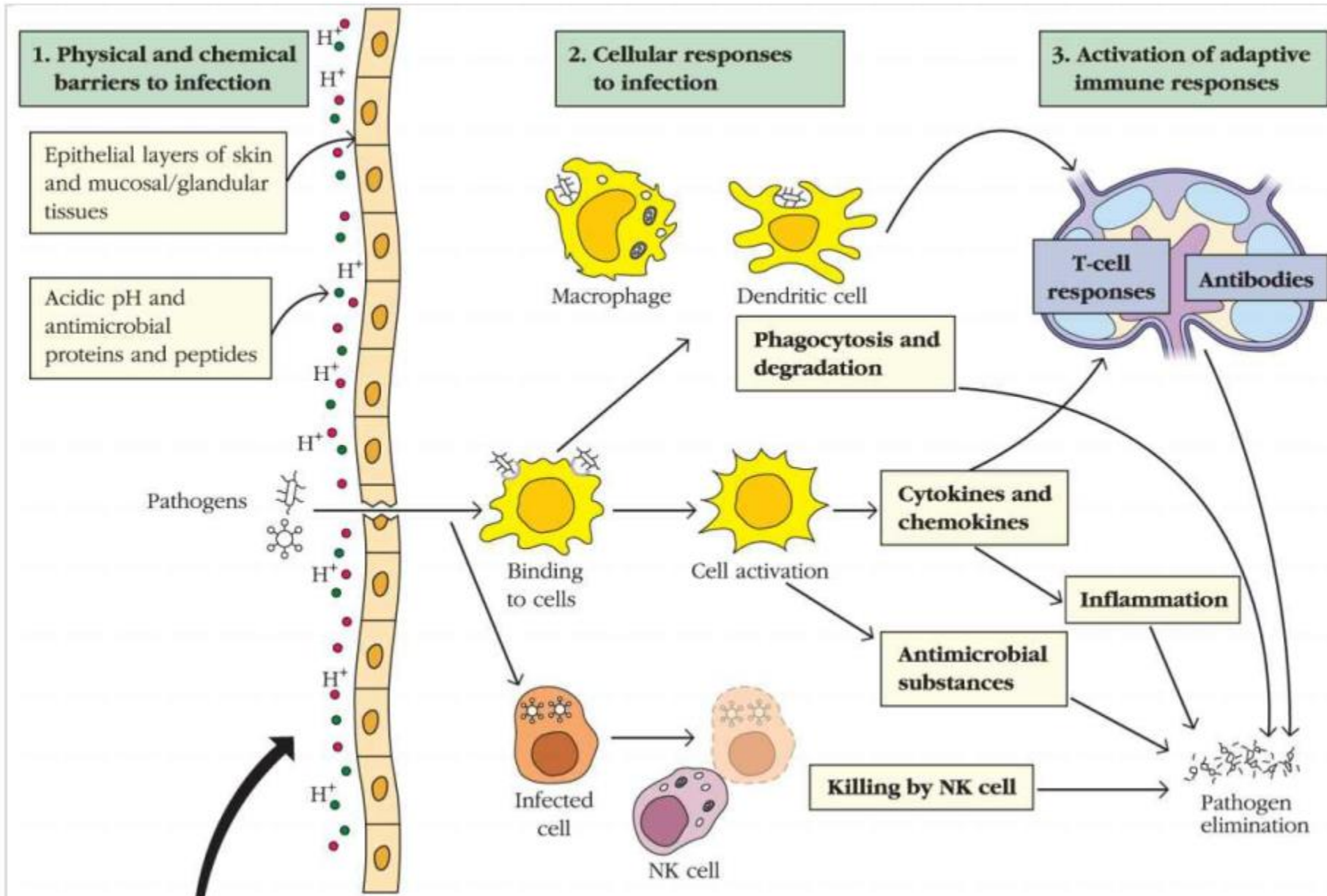
Innate immunity ...

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- The innate immune system includes
 - A. anatomical barriers (physical and chemical)
 - Physical barriers
 - the body's first line of defense
 - are the epithelial layers of the skin and mucosal surfaces
 - these epithelial barriers prevent infection by blocking pathogens from entering the body
 - Chemical barriers at these surfaces
 - include specialized soluble substances that possess antimicrobial activity as well as acid pH
 - B. Cellular responses
 - constitute the innate immune system's 2nd line of defense

Innate Immunity...

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Anatomical Barriers

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- The most obvious components of innate immunity
- Epithelial layers
 - *Skin and mucosal membranes*
 - Prevent pathogen entry into the body's interior
- Secretions (mucus, urine, saliva, tears, and milk)
 - wash away potential invaders
- Mucus
 - the viscous fluid secreted by specialized cells of the mucosal epithelial layers
 - entraps foreign microorganisms
- Cilia
 - The synchronous movement of cilia propels mucus-entrapped microorganisms from these tracts
- Coughing
 - is a mechanical response that helps us get rid of excess mucus

Chemical barriers

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- **Lysozyme:** found in saliva, tears, and fluids of the respiratory tract
 - cleaves the peptidoglycan components of bacterial cell walls
- **Complement:** Lyse pathogens or facilitate phagocytosis
- **Stomach acidity:** Kill most ingested pathogens
- **Mucus and acidic pH of vagina:** provide protection against bacterial and fungal pathogens
- **Lactoferrin:** Bind and sequester metal ions needed by bacteria and fungi, limiting their growth
- **Defensins (α and β):** Disrupt membranes
- **Commensal microbiota:** provide a first defense against pathogens
 - By competing with pathogens for nutrients and epithelial attachment sites
 - Maintenance of acidic pH in the vagina by the release of lactic acid by commensal lactobacteria

Cellular innate immune responses

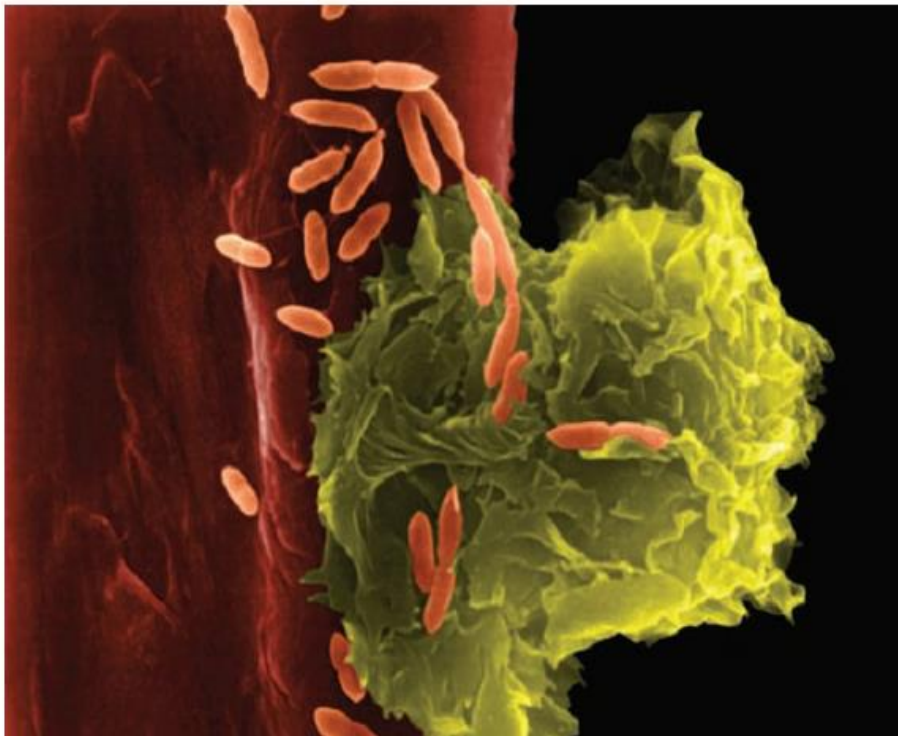
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- Pathogens can evade epithelial layers
 - Some pathogens have evolved strategies to penetrate epithelial layers
 - Epithelia may be disrupted by wounds, abrasions, and insect bites
- Once pathogens penetrate the epithelial barrier, an array of cellular membrane receptors and soluble proteins play the essential roles
- Phagocytic cells make up the next line of defense
 - Macrophages, Neutrophils, Dendritic cells, Monocytes
- Recruitment of fluid, cells, and molecules to sites of infection:
Inflammation

Phagocytosis...

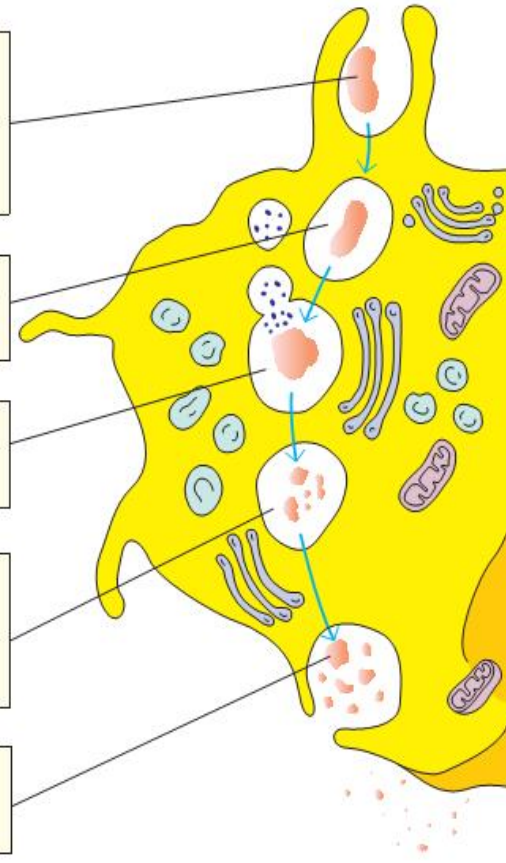
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(a)



(b)

- 1 Bacterium becomes attached to membrane evaginations called pseudopodia.
- 2 Bacterium is ingested, forming phagosome.
- 3 Phagosome fuses with lysosome.
- 4 Bacterium is killed and then digested by lysosomal enzymes.
- 5 Digestion products are released from cell.



How do phagocytic cells recognize microbes?

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- Using Pattern Recognition Receptors (PRRs)
- PRRs
 - Conserved variety of receptors
 - Recognize Pathogen-associated molecular patterns (PAMPs) → activate innate immune responses
 - Critical in the first line of defense against invaders
 - E.g. TLRs, Nod-like receptors (NLRs), C-type lectin receptor (CLR), Retinoic acid-inducible gene-I-like receptors (RLRs)
- PAMPs
 - unique antigenic structures expressed by pathogens but not by mammalian hosts
 - E.g. LPS, Peptidoglycan, dsRNA, ...

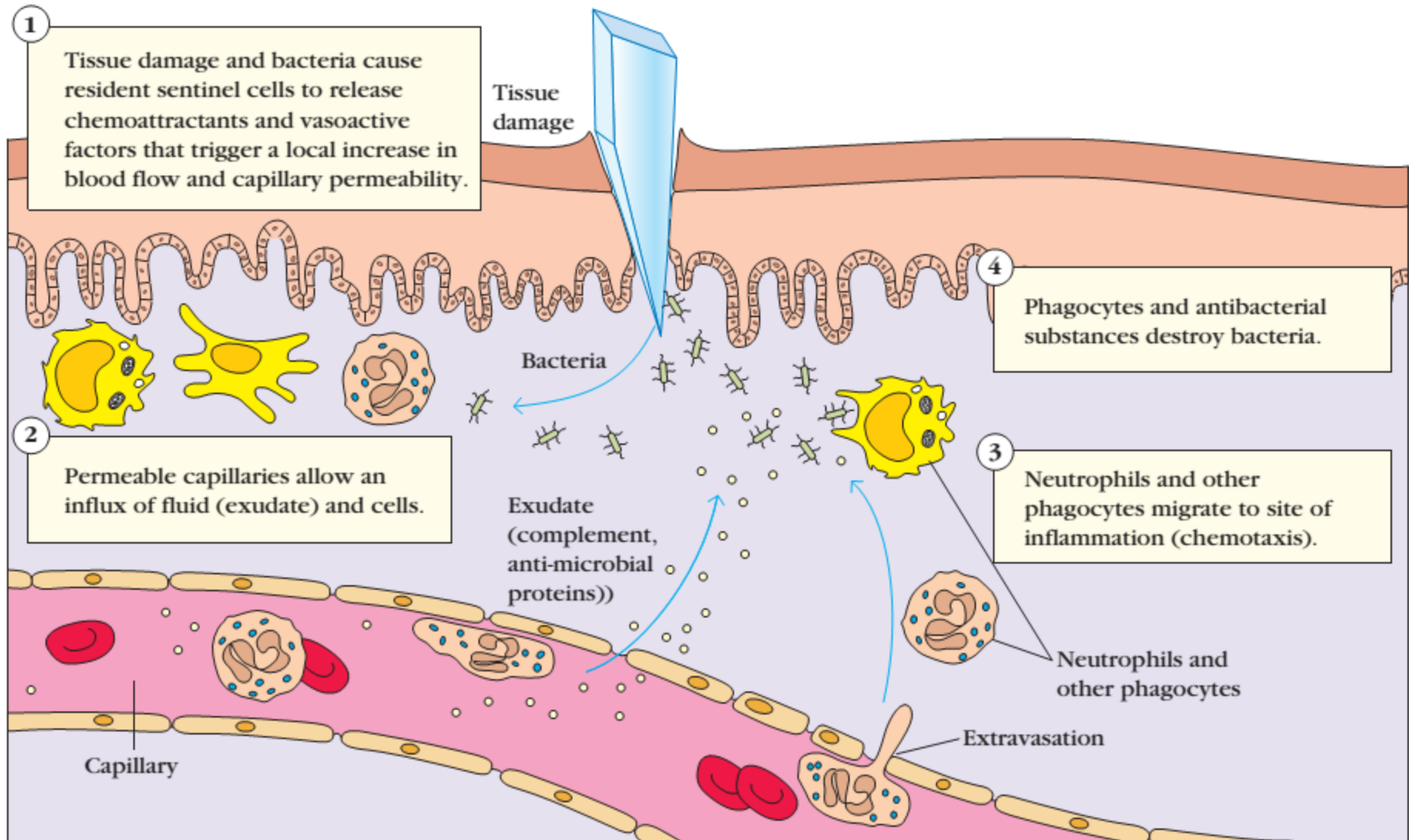
Inflammatory Responses

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- Inflammatory responses are initiated by innate immune responses to local infection or tissue damage
 - in particular, by the pro-inflammatory cytokines IL-1 β , TNF, and IL-6
- inflammatory responses
 - increased vascular permeability $\rightarrow$ allowing soluble innate mediators to reach the infected or damaged site
 - recruitment of neutrophils and monocytes from the blood into the site through the action of chemokines
- Local innate and inflammatory responses usually are beneficial in that
 - they eliminate pathogens and damaged or dead cells
 - promote healing
 - help to activate adaptive immune responses

Inflammatory Responses...

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Overview of immune system

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Adaptive immunity

- occurs after exposure to an agent, improves upon repeated exposure, and is specific
- mediated by B lymphocytes (development mainly occurs in the bone marrow) and by T lymphocytes (development mainly occurs in the thymus)
- Unlike innate immune cells, T cells and B cells recognize antigens, rather than universal microbial “patterns.”

Overview of immune system

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Adaptive immunity...

- B and T lymphocytes share three important features
 1. They exhibit remarkable diversity (i.e., collectively they can respond to millions of different antigens)
 2. they have a long memory (i.e., they can respond many years after the initial exposure because memory T cells and memory B cells are produced)
 3. they exhibit exquisite specificity (i.e., their actions are specifically directed against the antigen that initiated the response)

Overview of immune system

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Adaptive Immunity

- Does not develop until there is an antigenic challenge to an organism
- Mostly develops within five or six days after the initial exposure to an antigen
- Responds to the challenge with a high degree of specificity and “memory”
- Additional characteristics: Diversity, Self/non-self recognition
- Has two arms: Humoral and cell-mediated

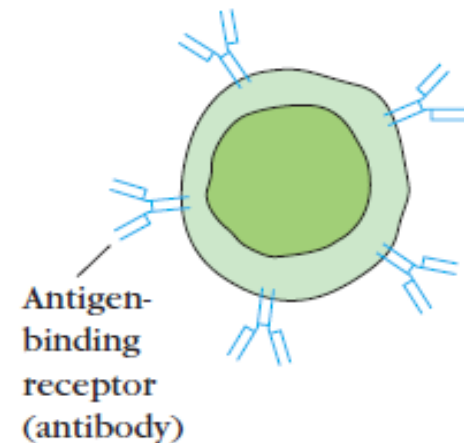
Overview of immune system

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Humoral immunity

- B cells are responsible by secreting antibody
- Antibody functions as the effector of the humoral response by binding to antigen
- When B-cells leave the BM, each expresses a unique antigen-binding receptor on their membrane (B-cell receptor)
- B-cell receptor = membrane-bound **antibody molecule**

(a) B cell

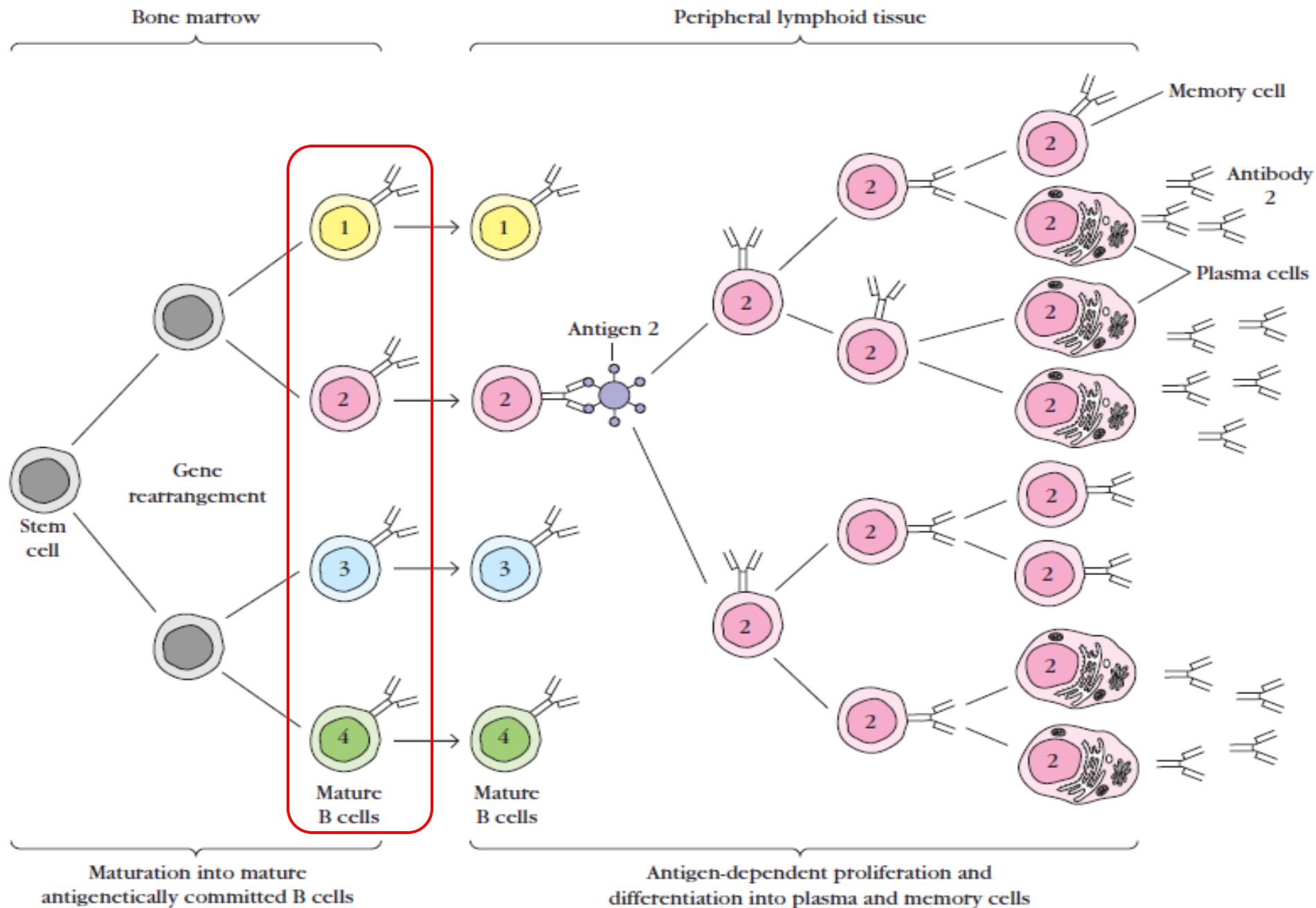


B lymphocytes...

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- Binding of naive B cell to an antigen causes the cell to divide rapidly and differentiated into **memory B cells** and **effector B cells** called **plasma cells**
- Memory B cells have a longer life span than naive cells, and they express the same membrane-bound antibody as their parent B cell
- Plasma cells produce the antibody in a form that can be secreted and have little or no membrane-bound antibody

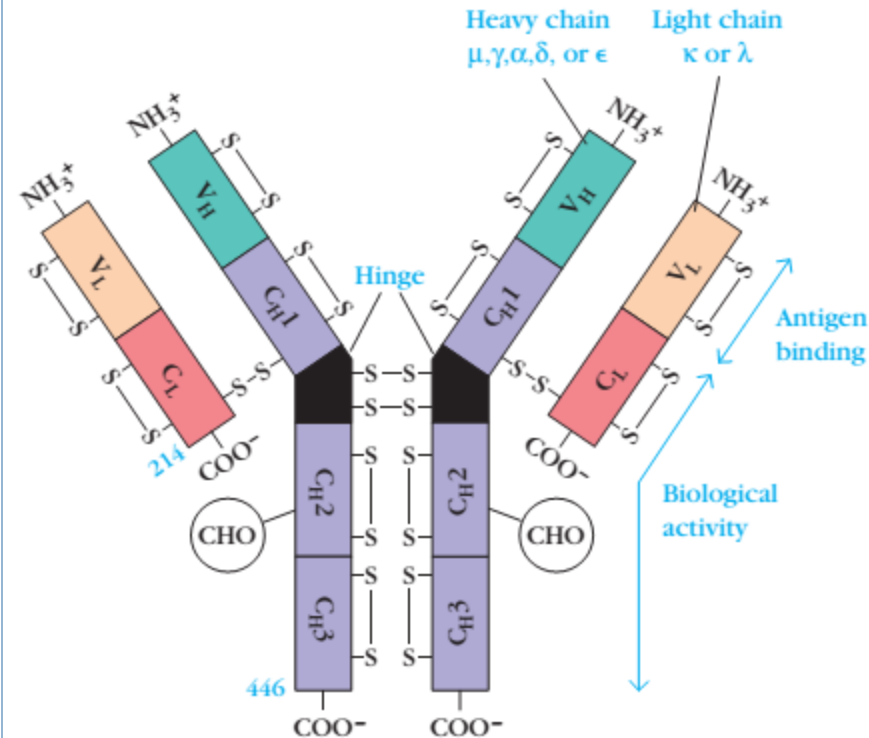
Maturation and clonal selection of B lymphocytes



Antibody

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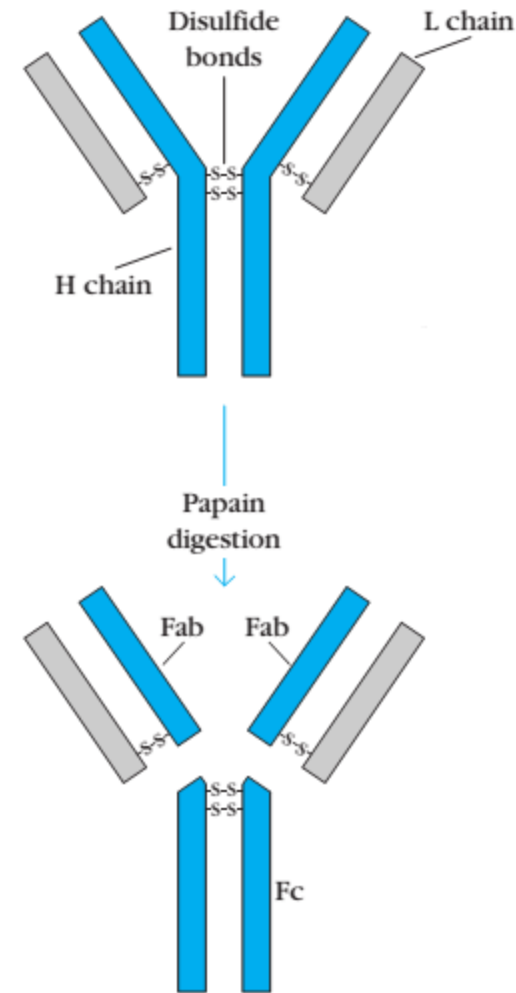
- Antigen binding protein present on the B-cell membrane and secreted by plasma cells
- Antibody molecules have a common structure of four peptide chains
 - two identical light (L) chains and two identical heavy (H) chains
- Each light chain & heavy chain, and the two identical heavy chains are bound by disulfide bond
- Each heavy and light chain contains
 - amino-terminal variable (V) region
 - constant (C) region: defines the two light-chain subtypes (κ , λ) and the five heavy-chain subclasses (μ , δ , γ , ϵ , α)



Antibody...

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- Most differences among antibodies fall within areas of the **V regions** called **complementarity determining regions (CDRs)** → **Antigen binding site**
- Immunoglobulins possess Fab and Fc Regions
 - **Fab** (“fragment, antigen binding”)
 - **Fc** fragment (“fragment, crystallizable”)



Antibody...

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- Each of the five different heavy chains is called an **isotype**
- The heavy chains of a given antibody molecule determine the class of that antibody: **IgM (μ)**, **IgG(γ)**, **IgA(α)**, **IgD(δ)**, or **IgE(ϵ)**
- Minor differences in the amino acid sequences of the γ and α heavy chains led to further classification of the heavy chains into **subisotypes**

Class	Heavy chain	subclass	Light chain
IgG	γ	$\gamma 1, \gamma 2, \gamma 3, \gamma 4$	κ or λ
IgM	μ	None	κ or λ
IgA	α	$\alpha 1, \alpha 2$	κ or λ
IgE	ϵ	None	κ or λ
IgD	δ	None	κ or λ

Antibody...

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Immunoglobulin G (IgG)

- The most abundant class in serum (80% of serum Ig)
- IgG readily cross the placenta and play an important role in protecting the developing fetus
- Can activate complement and involves in opsonization

Immunoglobulin A (IgA)

- Constitutes 10%–15% of the total immunoglobulin in serum
- Predominant Ig class in external secretions such as breast milk, saliva, tears, and mucus
- The daily production of secretory IgA is greater than that of any other immunoglobulin class
- IgA binds tightly to **poly-Ig receptor**, expressed on the surface of most mucosal epithelia

Antibody...

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Immunoglobulin M (IgM)

- accounts for 5%–10% of the total serum Ig
- Is pentameric; i.e. 10 Ag binding sites
- the first Ig class produced in a primary response to an antigen
- is the first to be synthesized by a neonate
- More efficient than other isotypes
 - in binding antigens
 - at activating complement
- Because of its large size, IgM does not diffuse well
 - found in very low concentrations in the intercellular tissue fluids

Antibody...

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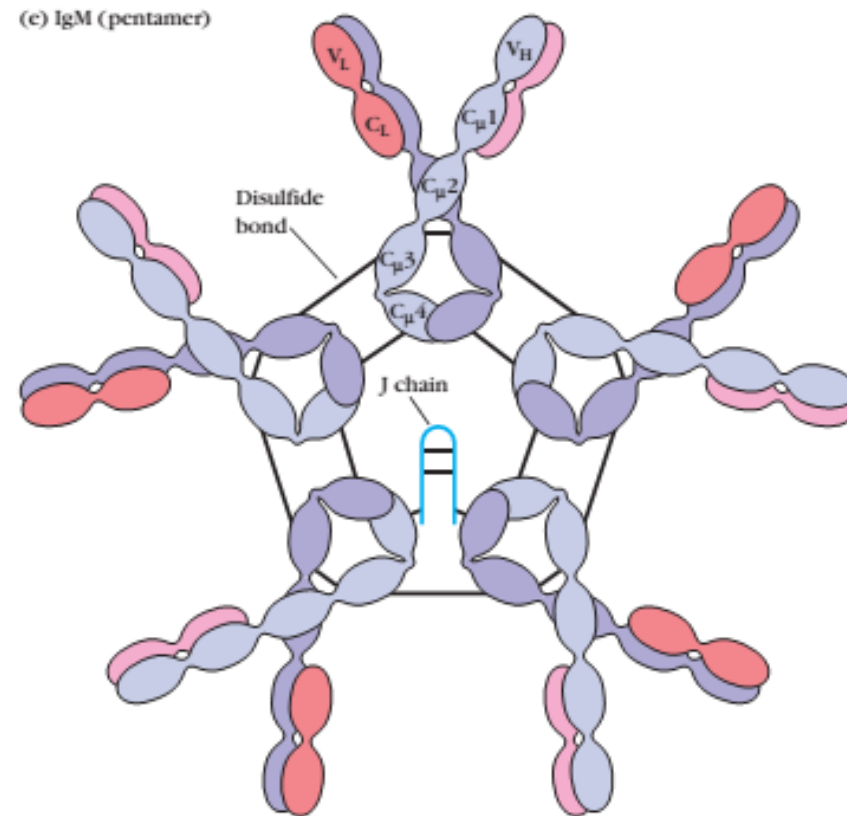
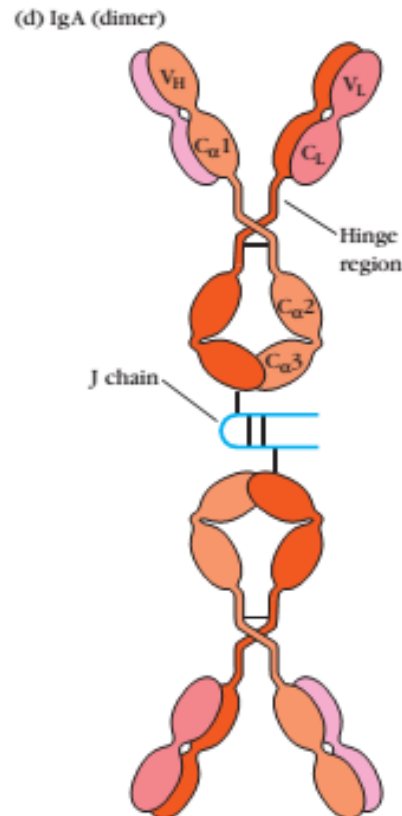
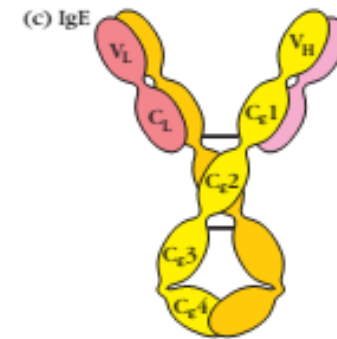
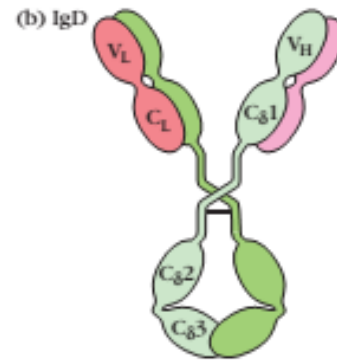
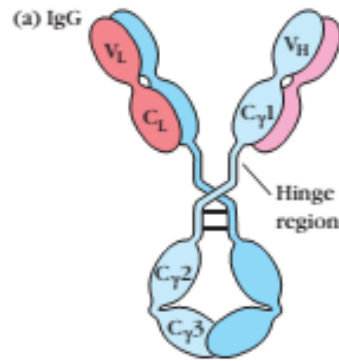
Immunoglobulin E (IgE)

- Constitutes $<0.001\%$ of total serum immunoglobulins
- IgE binds to Fc receptors on the membranes of blood basophils and tissue mast cells
 - mediate Type I hypersensitivity reactions
- Also plays important role against parasitic infection

Immunoglobulin D (IgD)

- constitutes about 0.2% of the total immunoglobulin in serum
- IgD, together with IgM, is the major membrane bound immunoglobulin expressed by mature B cells
- Its effector function isn't identified yet

General structures of the five major classes of secreted antibody



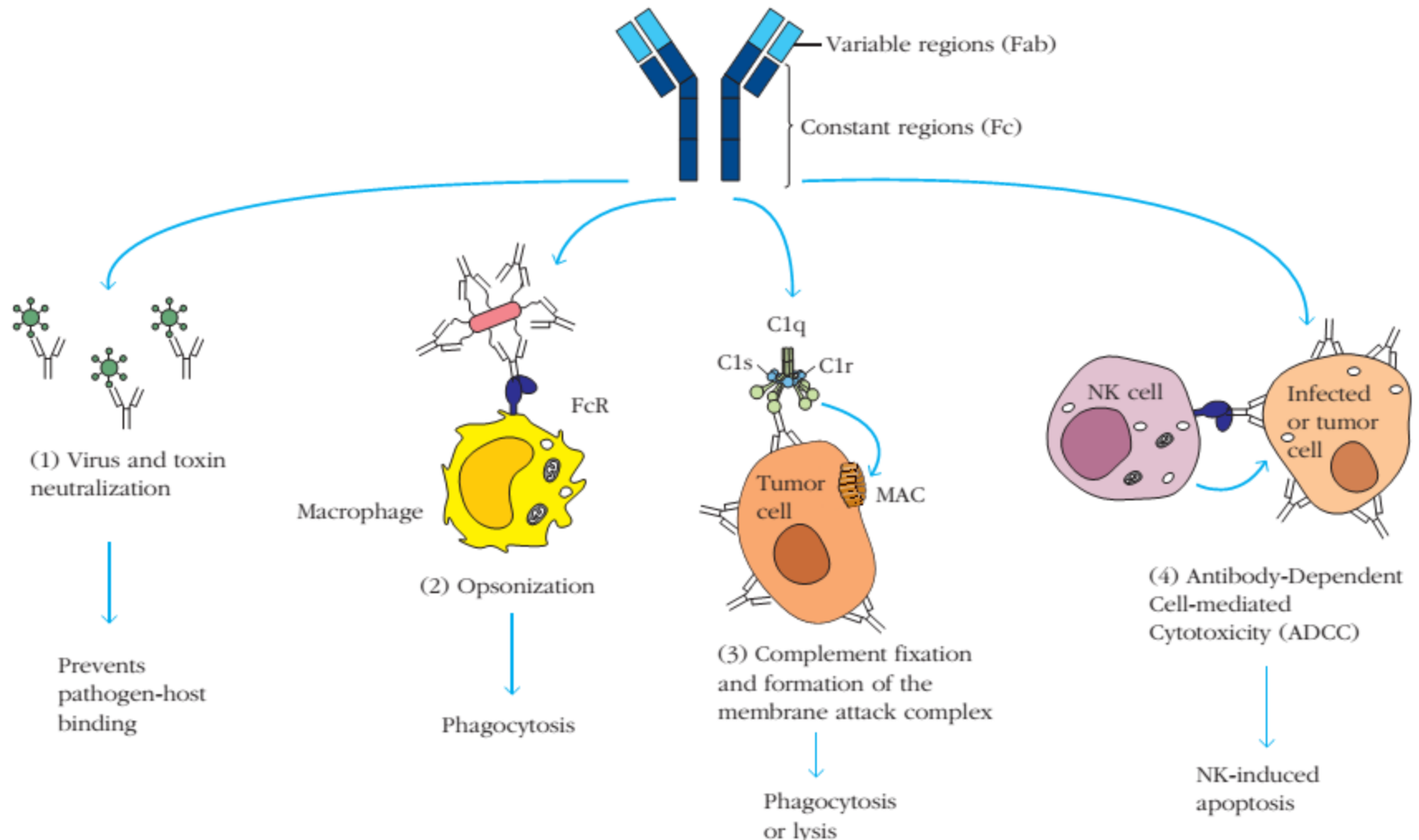
Antibody...

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- B/c antibodies are glycoproteins, they can function as potent immunogens to induce an antibody response
- **Polyclonal antibodies:** heterogeneous, comprising a mixture of antibodies, each specific for one epitope
- **Monoclonal antibodies:** derived from a single B-cell clone and thus specific for a single epitope (used for research, diagnostic, and therapeutic purposes)

Antibody-Mediated Effector Functions

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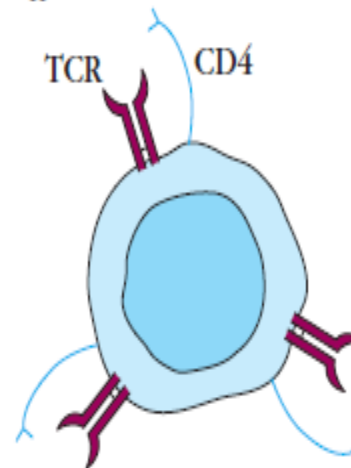
Overview of immune system

62

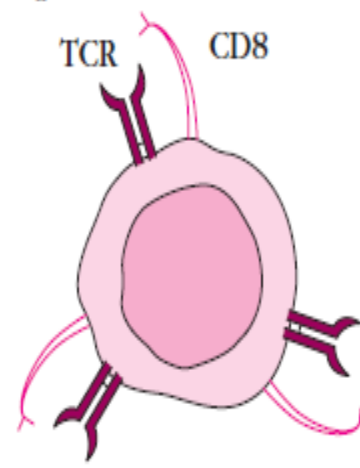
Cell mediated immunity

- **T Lymphocytes** arise in the bone marrow → migrate to the thymus gland to mature
- During their maturation T cells express a unique antigen-binding molecule, called the **T-cell receptor**, on their membrane

(b) T_H cell



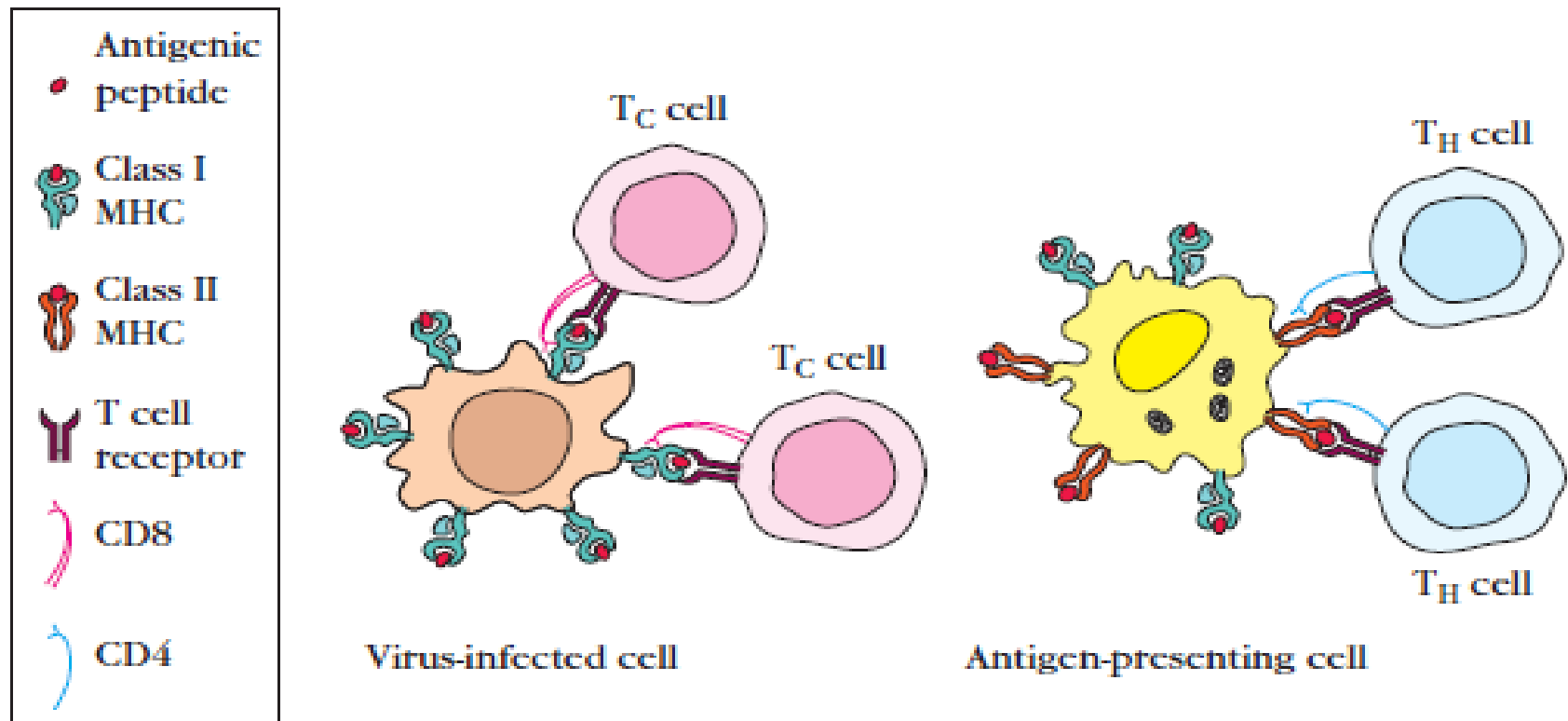
(c) T_C cell



Overview of immune system

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- T-cell receptors recognize antigen bound to **Major Histocompatibility Complex (MHC)** molecules



Class I MHC Molecules

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- **MHC I molecules are present on all nucleated cells in the body**
- **Specialized in presenting antigens that originate from the cytosol, such as viral proteins**
- **Present Ag to CD8⁺ T cells, which recognize and kill cells expressing such intracellular antigens**
- **The MHC I peptide-binding groove can bind a peptide of 8 to 10 amino acids**

Class II MHC Molecules

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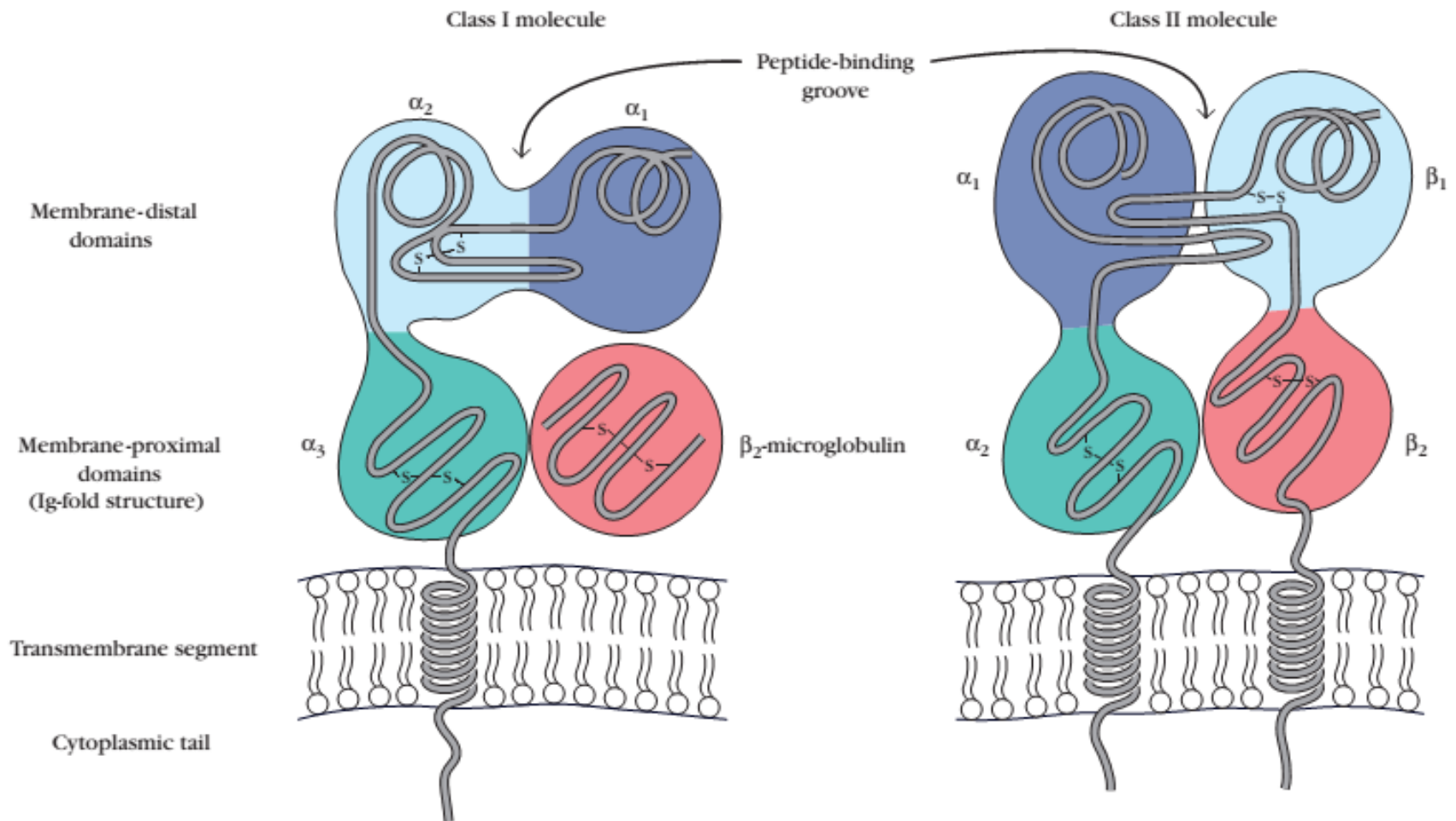
- Expressed almost exclusively on antigen-presenting cells (APCs)
- Specialize in presenting antigens from extracellular spaces that have been engulfed by APCs
- Presents the antigenic peptide to CD4⁺ T cells
- The peptide-binding groove of MHC II can bind peptides ranging from 13-18 amino acids

TABLE 8-4 Antigen-presenting cells

Professional antigen-presenting cells	Nonprofessional antigen-presenting cells	
Dendritic cells (several types)	Fibroblasts (skin)	Thymic epithelial cells
Macrophages	Glial cells (brain)	Thyroid epithelial cells
B cells	Pancreatic beta cells	Vascular endothelial cells

MHC structure

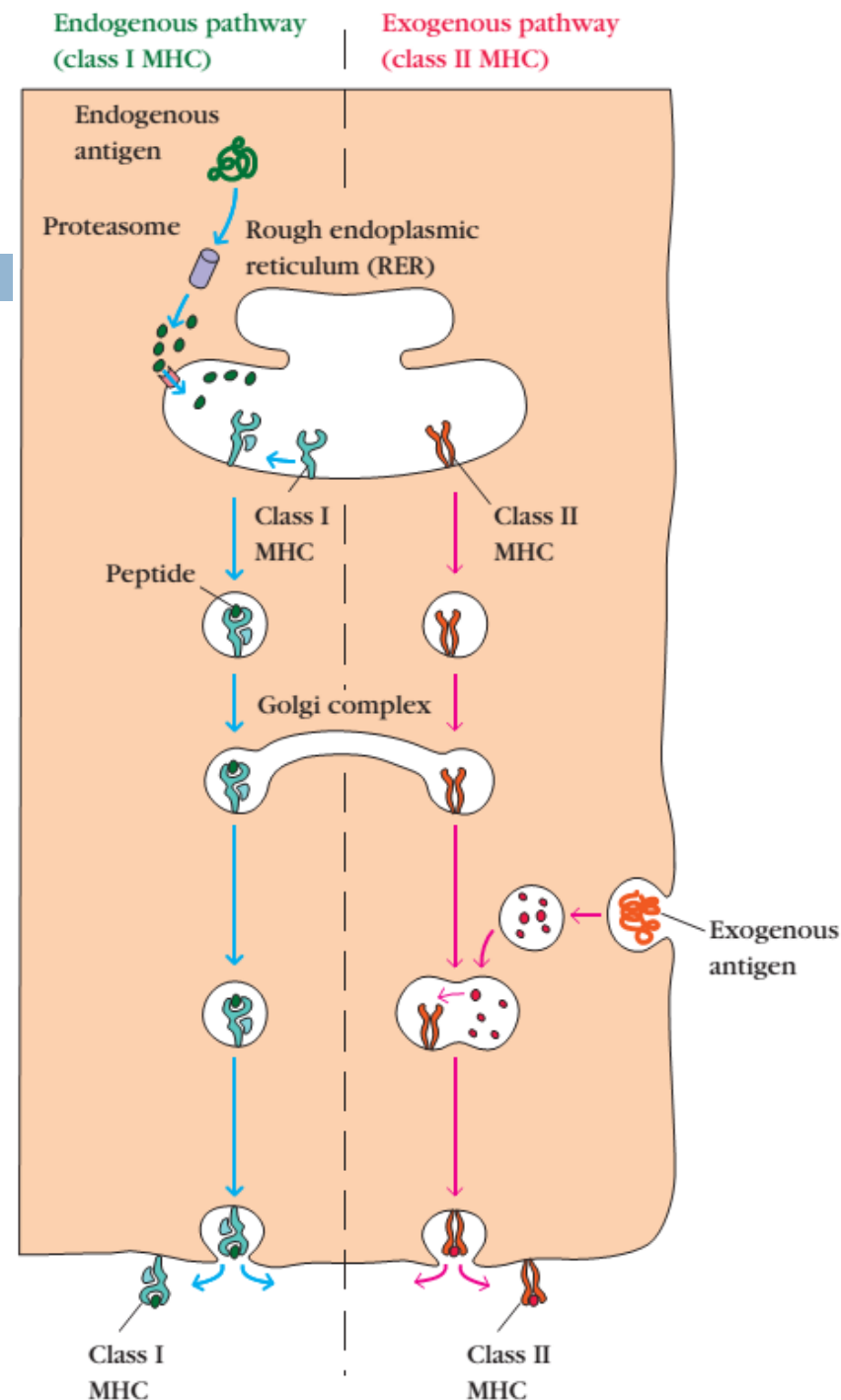
66



Antigen Processing and Presentation Pathways

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- The immune system uses d/t pathways to eliminate intracellular and extracellular antigens
- Endogenous antigens
 - generated within the cell
 - processed in the **cytosolic or endogenous pathway**
 - presented on the membrane with MHC I
- Exogenous antigens
 - taken up from the extracellular environment by endocytosis
 - typically processed in the **exogenous pathway**
 - presented on the membrane with MHC II

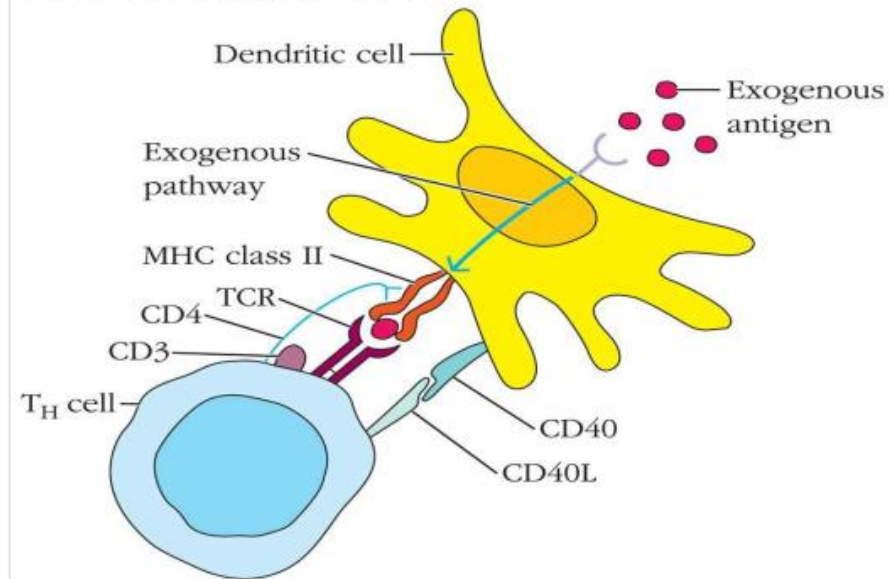


Cross-Presentation

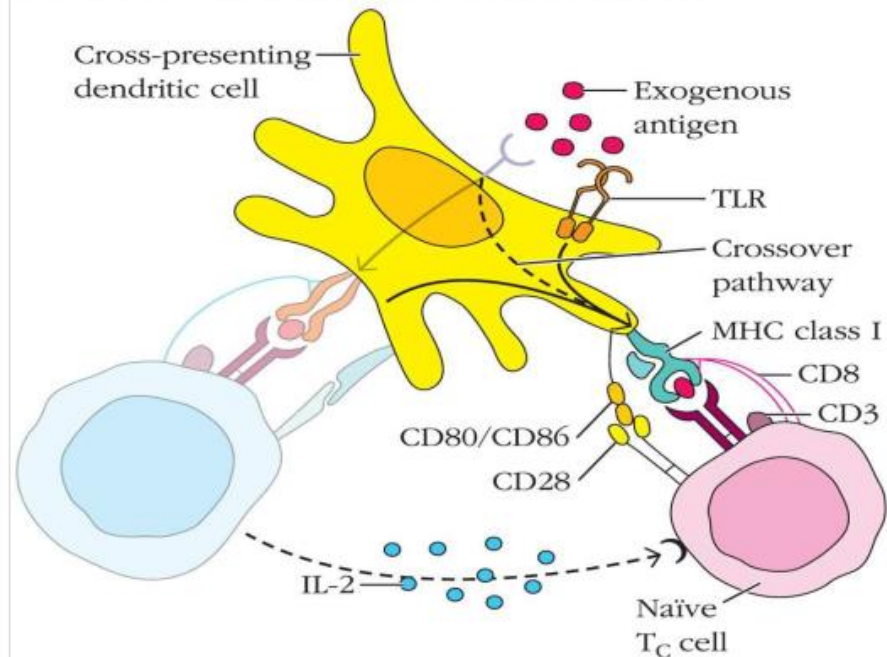
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- Exogenous antigens internalized by dendritic cells can gain access to the endogenous presentation pathway leading to
 - peptide association with MHC I and engagement with CD8 T cells
- allows these APCs to capture antigen, such as viral proteins, from the extracellular environment or from dying cells, process these antigens, and activate CTLs that can then seek out and attack virus-infected cells, inhibiting further spread of the infection

(a) DC licensing by T_H cell



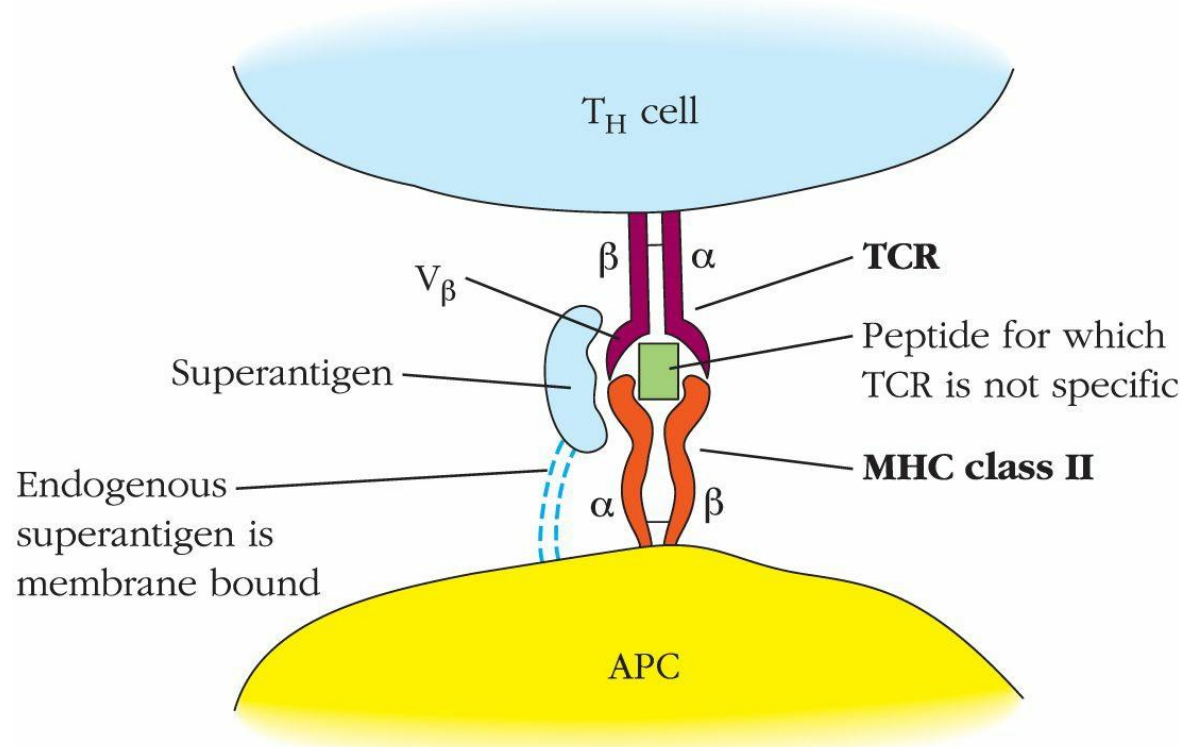
(b) DC cross-presentation and activation of CTL



Reading assignment

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- **Superantigens**



Punt, Kuby Immunology, 8e, © 2018 W. H. Freeman and Company

T-cells response

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- ❖ When a naive T cell encounters antigen combined with a MHC molecule on a cell, it proliferates and differentiates into memory T cells and various effector T cells



Overview of immune system

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Activated TH cells

- secrete various cytokines
 - activate B cells, TC cells, Macrophages
 - various other cells that participate in the immune response
- Important against extracellular pathogens

Activated CTLs

- Play an important role in the killing of virus infected cells and tumor cells
- initiate killing of their targets via the delivery of pro-apoptotic molecules
 - Perforin
 - Granzymes (fragmentins)

Innate and adaptive immune responses work together to eliminate foreign invaders

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- **The phagocytic cells crucial to nonspecific immune responses are intimately involved in activating the specific immune response**
- **various soluble factors (cytokines) produced by a specific immune response enhance the activity of phagocytic cells**
- **The specific immune response serve to regulate the intensity of the inflammatory response**

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The complement system

The complement system

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- **Complement**
 - refers to a set of >50 serum proteins
 - cooperates with both innate and adaptive immune systems to eliminate pathogens, dying cells, and immune complexes from the body
 - first discovered by virtue of their ability to complex with antibodies and destroy the membranes of antibody-bound cells

The complement system

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- Most complement components are **synthesized** in the **liver** by hepatocytes
 - some are also produced by blood monocytes, tissue macrophages, fibroblasts, and epithelial cells of the gastrointestinal and genitourinary tracts
- Complement components constitute ~15% of the globulin protein fraction in plasma

Major pathways of complement activation

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1. The Classical Pathway

- initiated when C1q binds to antigen-antibody complexes

2. Lectin pathway

- Initiated when mannose-binding lectin (MBL) binds specifically to conserved carbohydrate arrays on pathogens

3. Alternative pathway

- presence of a pathogen alone

Major pathways of complement activation

Classical pathway

Antigen-antibody
immune complexes

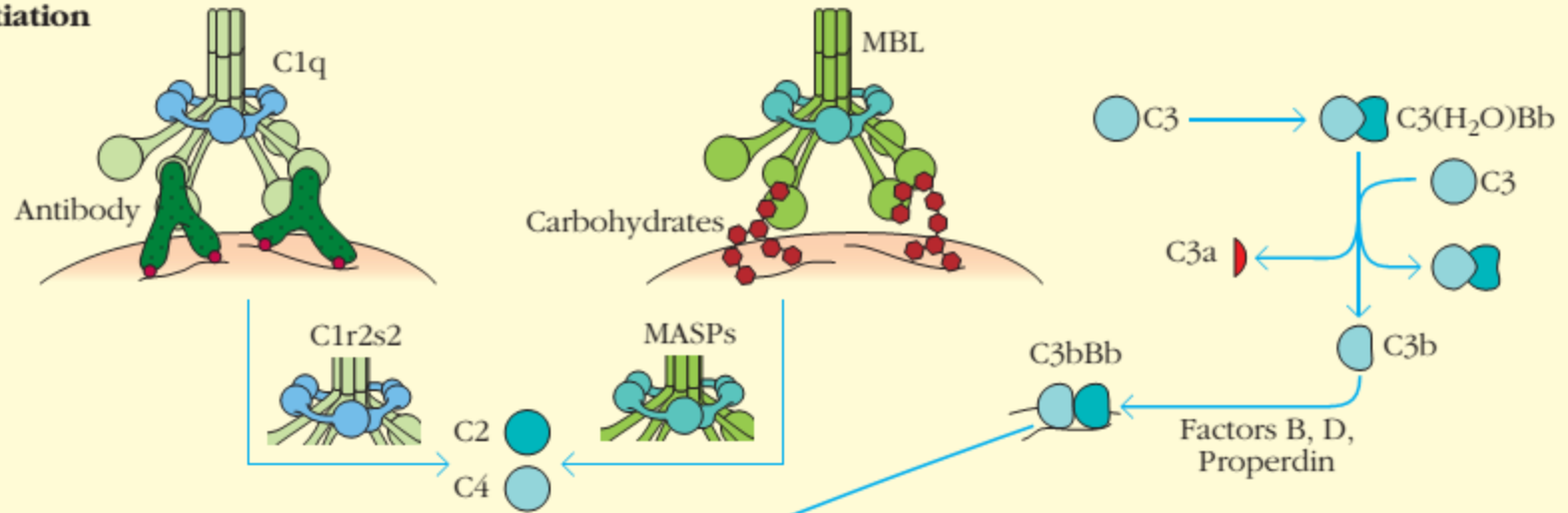
Lectin pathway

PAMP recognition
by lectins

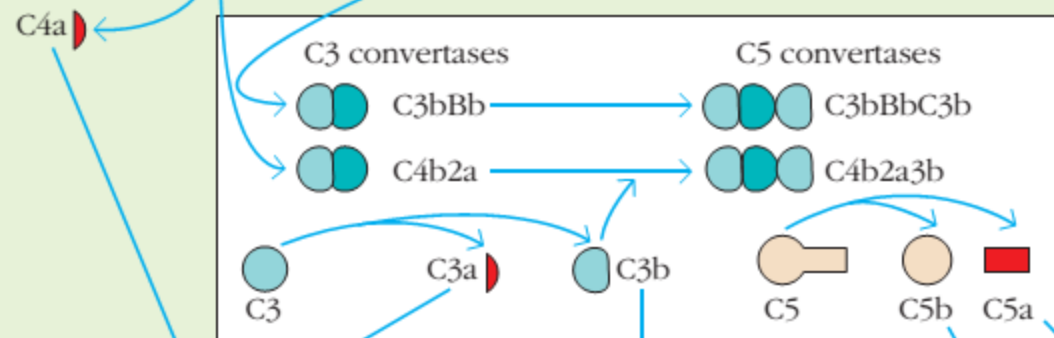
Alternative pathway

Spontaneous hydrolysis
or pathogenic surfaces

Initiation



Amplification



Termination

Inflammation

Opsonization

Lysis

Inflammation

Complement functions

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1. C3a and C5a recruit phagocytic cells to the site of infection and promote inflammation
2. C3b bound to a pathogen acts as an opsonin
 - enabling phagocytes that express receptors for C3b to ingest the complement-coated microbe more easily
3. Formation of a membrane-attack complex (MAC) disrupts cell membrane and causes cell lysis

Immunization

Immunization is the process of eliciting a state of protective immunity against a disease-causing pathogen

- Passive Immunization
- Active Immunization

Passive Immunization

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- Involves transfer of preformed Antibodies
- generates no memory response
- protection is transient
- **Naturally acquired:** by transfer of preformed maternal antibodies across the placenta to a developing fetus or in colostrum and milk to an infant
- **Artificially acquired:** by injecting a recipient with preformed antibodies
 - Routinely administered to individuals exposed to botulism, tetanus, diphtheria, and rabies

Active immunization

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- The goal of active immunization is to elicit protective immunity and immunologic memory
- When active immunization is successful, a subsequent exposure to the pathogenic agent elicits a heightened immune response that successfully eliminates the pathogen or prevents disease mediated by its products
- Can be achieved by
 - Natural infection with a microorganism or
 - Artificially by administration of a **vaccine**

Vaccine

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Types

- Live attenuated vaccines
- Inactivated/ Killed vaccines
- Subunit vaccines
- Recombinant-Vector Vaccines
- DNA vaccines
- mRNA vaccines

Vaccine

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Live attenuated vaccines

- Pathogens are attenuated (lose their pathogenicity) but retain their capacity for transient growth within an inoculated host
- E.g. Measles, mumps, rubella, oral polio, smallpox, varicella and yellow fever
- Advantages
 - Their capacity for transient growth results in increased immunogenicity and production of memory cells
 - Often require only a single immunization
 - Particularly suitable for inducing a cell-mediated immune response
- Drawbacks
 - Possibility of reversion to a virulent form
 - Requires refrigerated storage

Vaccine...

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Inactivated/ Killed vaccines

- Virulent pathogen is inactivated usually with formaldehyde
- Example: Pertussis, Anthrax , Polio (Salk), Rabies
- Advantage
 - Safer and more stable than attenuated vaccines
- Drawbacks
 - often require repeated boosters to maintain the immune status
 - induce a predominantly humoral antibody response
 - less effective in eliciting a secretory IgA response
 - potential risk for incomplete inactivation

Vaccine...

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Subunit vaccines

- Consist of specific, purified macromolecules derived from pathogens
- Advantage
 - Suitable if either a safe attenuated strain is unobtainable or for uncultivable pathogens
 - Safer than attenuated or inactivated vaccines
- Drawbacks
 - traditionally less immunogenic than attenuated or inactivated vaccines
 - require larger and more frequent doses of antigen, as well as co-delivery of adjuvants

Vaccine	component
<i>S. Pneumoniae</i>	Capsular polysaccharides
<i>N. meningitidis</i>	
<i>H. influenzae type b (Hib)</i>	
HPV, HBV	Virus like particle (VLP): mimic viral structure but no genetic material
Diphtheria Tetanus	Toxoid (formaldehyde inactivated toxin)

Vaccine...

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Recombinant-Vector Vaccines

- Genes encoding major antigens of a pathogen are introduced into attenuated viruses or bacteria (E.g. vaccinia virus, Adenovirus, BCG strain of *M. bovis*): E.g. SARS-CoV-2 vaccine (Astrazenca)
- Drawbacks
 - Prior immunity to the vector can prevent vaccine efficacy
 - Safety concern with the vector

DNA Vaccines

- plasmid DNA encoding antigenic proteins is injected directly into the muscle of the recipient
- Muscle cells take up the DNA and the encoded protein antigen is expressed, leading to both humoral and a cell-mediated response

mRNA vaccines

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- mRNA vaccines make proteins in order to trigger an immune response
- mRNA vaccines have several benefits compared to other types of vaccines
 - shorter manufacturing times
 - because they do not contain a live virus, no risk of causing disease in the person getting vaccinated
 - Example: SARS-CoV-2 vaccines: Pfizer, Moderna, Janssen

Vaccine...

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Herd immunity

- Vaccination is not 100% effective; small percentage of recipients will respond poorly and therefore will not be adequately protected
- If the majority of the population is immune to an infectious agent, the chance of a susceptible individual contacting an infected individual is so low that the susceptible one is not likely to become infected
- This phenomenon is known as **herd immunity**

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Failures of the immune response

Microbes immune evasion mechanisms

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- **Pathogens use different mechanisms to escape from immune destruction**
 - Antigenic variation
 - Latency
 - Resist destruction by host defense mechanisms

Antigenic variation

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- Pathogens evade immune surveillance by altering their antigens
- allows pathogens to escape from immunity
- Particularly important for extracellular pathogens
- Antigenic variation can occur through three ways
 - Existence in a wide variety of antigenic types
 - Antigenic shift and drift
 - Programmed rearrangements in the DNA of the pathogen

Latency: Some pathogens persist in vivo by ceasing to replicate until immunity wanes

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- Some viruses can enter a state known as **latency** in which the virus is not being replicated
- In the latent state, the virus does not cause disease but, because there are no viral peptides to flag its presence, the virus cannot be eliminated
- Such latent infections can be reactivated resulting in recurrent illness
- Example: Herpesviruses

Some pathogens resist destruction by host defense mechanisms

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- Some pathogens induce a normal immune response but have evolved specialized mechanisms for resisting its effects
- *M. tuberculosis* is taken up by macrophages but prevents the fusion of the phagosome with the lysosome
- *L. monocytogenes* escape from phagosome into the cytoplasm of the macrophage, where they can multiply readily

Tolerance & Autoimmune Disease

Tolerance

95

- **Random gene rearrangements** during lymphocyte development inevitably result in the generation of **some lymphocytes with affinity for self antigens**
- How does the immune system manage these self reactive lymphocytes?
 - By generating self-tolerance
- **Self tolerance:** an individual's immune system doesn't recognize and attack the normal tissues of the body
- **Immune tolerance:** occurs when an immunocompetent host fails to respond to an immunogenic challenge with a specific antigen

Tolerance

96

- Successful maintenance of tolerance ensures that the host always knows the difference between self and non-self
- How does self tolerance ensured?
 - **Central tolerance:** occurs during lymphocyte development
 - **Peripheral tolerance:** occurs after lymphocytes leave the primary lymphoid organs

Peripheral tolerance

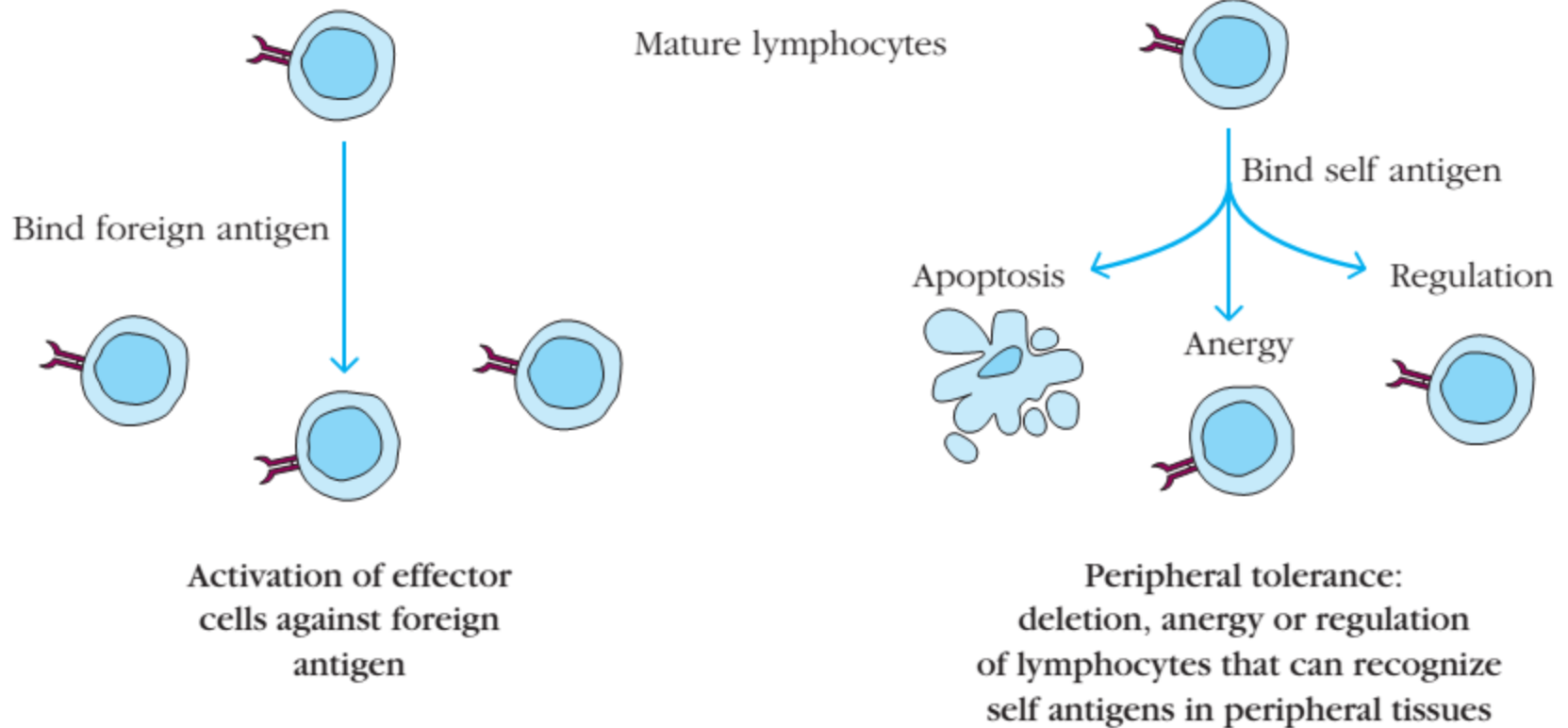
97

- Because central tolerance is not perfect
 - some self-reactive lymphocytes find their way into the periphery and secondary lymphoid tissues
- Additional safeguards limit their activity: peripheral tolerance
- Peripheral tolerance
 - Inactivate some self-reactive lymphocytes in secondary lymphoid tissues
 - Generates other lymphocytes (T_{REG}) that actively inhibit immune responses against self

Peripheral tolerance

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Peripheral tolerance



Autoimmunity

Failure or break in self-tolerance results in autoimmunity

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- Autoimmune disease occurs when a specific adaptive immune response is mounted against self antigens resulting in organs or tissues damage
 - loss or breakdown of self-tolerance
- When an adaptive immune response develops against self antigens, it is usually impossible to eliminate the antigen completely, and so a sustained response occurs

Autoimmunity

100

- **Genetic Factors**
 - Many auto immune diseases are more common in females than in males
 - Some autoimmune diseases run in families, suggesting a role for genetic susceptibility
 - if one of two identical twins is affected, then the other twin is quite likely to be as well
- **Hormonal Factors**
 - Approximately 90% of all autoimmune diseases occur in women
- **Environmental Factors**
 - Several environmental agents that trigger autoimmune diseases, most of which are either bacteria or viruses

Proposed Mechanisms for Induction of Autoimmunity

101

- **Molecular Mimicry**
- **Alteration of Normal Proteins**
- **Release of Sequestered Antigens**
- **Epitope Spreading**
- **Failure of Regulatory T Cells**

Some common autoimmune diseases

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Disease	Disease mechanism	Consequence	Prevalence
Psoriasis	Autoreactive T cells against skin-associated antigens	Inflammation of skin with formation of scaly patches or plaques	1 in 50
Rheumatoid arthritis	Autoreactive T cells against antigens of joint synovium	Joint inflammation and destruction causing arthritis	1 in 100
Graves' disease	Autoantibodies against the thyroid-stimulating-hormone receptor	Hyperthyroidism: overproduction of thyroid hormones	1 in 100
Hashimoto's thyroiditis	Autoantibodies and autoreactive T cells against thyroid antigens	Destruction of thyroid tissue leading to hypothyroidism: underproduction of thyroid hormones	1 in 200
Systemic lupus erythematosus	Autoantibodies and autoreactive T cells against DNA, chromatin proteins, and ubiquitous ribonucleoprotein antigens	Glomerulonephritis, vasculitis, rash	1 in 200
Sjögren's syndrome	Autoantibodies and autoreactive T cells against ribonucleoprotein antigens	Lymphocyte infiltration of exocrine glands, leading to dry eyes and/or dry mouth; other organs may be involved, leading to systemic disease	1 in 300
Crohn's disease	Autoreactive T cells against intestinal flora antigens	Intestinal inflammation and scarring	1 in 500
Multiple sclerosis	Autoreactive T cells against brain antigens	Formation of sclerotic plaques in brain with destruction of myelin sheaths surrounding nerve cell axons, leading to muscle weakness, ataxia, and other symptoms	1 in 700
Type 1 diabetes (insulin-dependent diabetes mellitus, IDDM)	Autoreactive T cells against pancreatic islet cell antigens	Destruction of pancreatic islet β cells leading to nonproduction of insulin	1 in 800

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Hypersensitivity

Hypersensitivity Reactions

- An immune response mobilizes a battery of effector molecules that induce a localized inflammatory response that eliminates antigen without extensively damaging the host's tissue
- Under certain circumstances, however, this inflammatory response can have deleterious effects, resulting in significant tissue damage or even death
 - This inappropriate immune response is termed **hypersensitivity** or **allergy**
- **Hypersensitivity:** harmful immune response that produce tissue injury and may cause serious disease

Classification of Hypersensitivity reaction

Based on

- The effector molecules
 - Type-I, II, III, IV
- Time required for manifestation of symptoms
 - Immediate hypersensitivity
 - Delayed-type hypersensitivity

END